

CLASS 06 / SEP 29 / U05

ATTRACTORS

P2 CONTINUES — THE RULE STOPS COMING FROM THE INDEX

NOTHING DUE

P2b ISSUED TODAY

AN ATTRACTOR IS A NUMBER

For every point: how far is it from something? Use that number to drive a property.

THE WHOLE IDEA

```
for each point p:  
    d = distance(p, attractor)  
    size = f(d)
```

That is it. Everything else
is a choice of f.

TWO NUMBERS, SAME JOB

INDEX told you where in the sequence.

DISTANCE tells you how far from something outside the
collection.

Both are numbers driving a property. That is the only idea this
week.

LAST WEEK YOU BUILT THE POSITIONS. THIS WEEK YOU MEASURE THEM.

READ THE FIELD BEFORE YOU CHANGE ANYTHING

Distance is a value at every point. Look at it as grey before it becomes geometry.

01 MEASURE

Every point gets a number: how far to the attractor.

02 DISPLAY

Show those numbers as grey.
This is a drawing already.

03 REMAP

Push that range into the range
your property needs.

04 DRIVE

Only now does the component
change. The field was finished
first.

THE FIELD, READ AS A GREY VALUE ACROSS THE PLANE



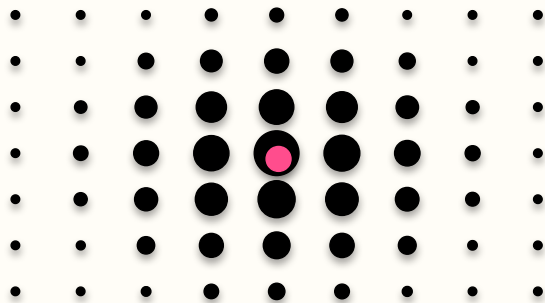
IF THE GREY IMAGE IS NOT ALREADY A GOOD DRAWING, THE COMPONENT WILL NOT SAVE IT.

P2b requires the measured field shown before the geometric result. That order is the assignment.

THREE KINDS

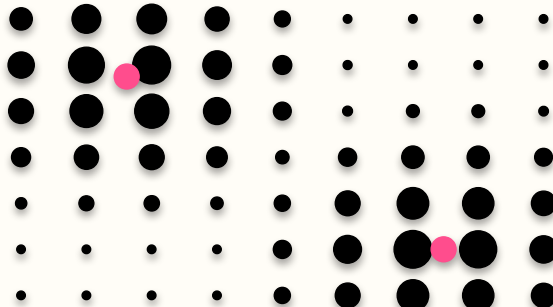
Same arithmetic. What changes is what you measure the distance to.

SINGLE POINT



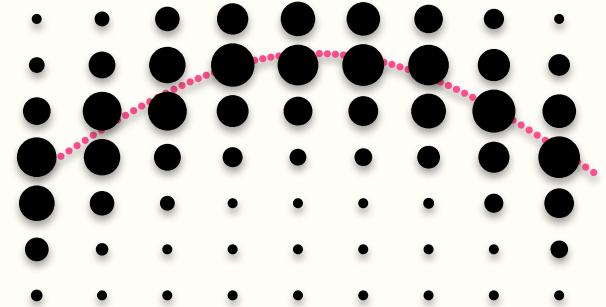
`distance(p, a)`

MULTIPLE POINTS



`nearest of several`

CURVE



`nearest location on a curve`

Whatever you measure to, the output is the same thing: one number per point. The attractor is not an object — it is a question you ask of every point.

REMAP, AND SAY WHERE YOU CLAMPED

Distance comes out in model units. Your property needs a specific range.

THE OPERATION

$$t = (d - d_{\min}) / (d_{\max} - d_{\min})$$
$$\text{out} = \text{out}_{\min} + t \times (\text{out}_{\max} - \text{out}_{\min})$$

d 0 ... 8.4 units ← source
out 0.1 ... 1.0 radius ← target

SUBTRACT. DIVIDE. MULTIPLY. ADD.

Nothing here is new — it is the arithmetic from Week 2 applied to a distance instead of a coordinate.

Clamping is a decision, not a safety net. Decide what happens past the end of your range and say so.

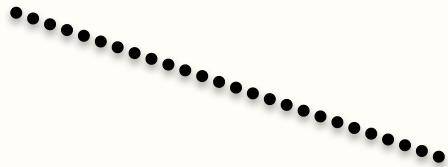
P2b REQUIRES SOURCE AND TARGET RANGES, CLAMPING AND FALLOFF STATED ON THE SHEET.

FALLOFF IS THE FUNCTION

f is your decision. It is where the graphic quality actually comes from.

LINEAR

$$v = d$$

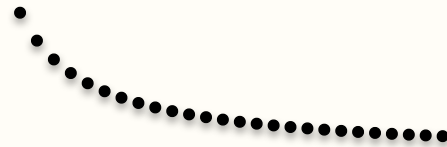


distance →

Even change all the way across.

INVERSE

$$v = 1 / d$$

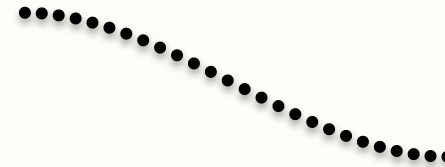


distance →

Sharp near the attractor, flat far off.

SMOOTH

eased



distance →

Soft at both ends. Usually the best-looking.

THRESHOLD

on / off at a cut



distance →

No gradient at all. Two populations.

Two students with the same attractor and different falloff produce completely different images. This is a design decision, not a setting.

THE SAME THING, WRITTEN OUT

Measure, remap, clamp, shape the falloff, then place. In that order.

```
for p in points:
    d = distance(p, attractor)          # or min(...) over
several [t = (d - d_min) / (d_max - d_min)] # remap
    t = clamp(t, 0, 1)
    t = falloff(t)                     # linear, inverse,
smooth, step size = out_min + t * (out_max - out_min)
    place(component, p, scale=size)
```

WHAT IT MAPS ONTO

distance
external condition
field
remap
clamp
falloff
nearest

LINE 2 IS THE ONLY LINE THAT CHANGES BETWEEN THE THREE ATTRACTOR TYPES.

ISSUED TODAY

P2b — ATTRACTOR FIELD STUDY

DUE WEEK 9 — Oct 20 / 7%

Hold the component and the arrangement constant, and change only what you measure against. The comparison is the assignment: three attractor types, one field logic, and an architectural intention you can defend.

COMPARE

THREE TYPES

Single point, multiple points, and a curve — same array.

SHOW

FIELD FIRST

The measured field, then the geometric result.

ARGUE

ONE INTENTION

A spatial or architectural reason, demonstrated live.

REQUIREMENTS — P2b

Visual quality 35% · computational understanding 25% · technical execution 20% · experimentation 10% · requirements and identity 10%.

01

On up to two sheets, compare single-point, multiple-point and curve attractors using the same component and arrangement.

02

Show the measured field before the geometric result. State source and target ranges, clamping and falloff.

03

Develop one spatial or architectural intention and demonstrate a live change to the attractor or mapping.

CHECKPOINT · WEEK 8 — Oct 13 — Bring the grayscale field and a first geometric response to midterm as work in progress.

NOW — YOUR FIRST FIELD

Rest of the session: measure, look, remap, and only then attach geometry.

✓

Distance to one point measured across your existing array

✓

The field displayed as grey, on its own, before anything changes shape

✓

Source and target ranges written down — actual numbers, not guesses

✓

One property driven: size, height or rotation. Just one, for now

✓

The attractor moved, and you can explain the response without looking at the graph

MOVE THE ATTRACTOR. IF YOU CANNOT PREDICT WHAT HAPPENS, STOP AND READ THE FIELD.

BEFORE NEXT CLASS

WATCH

U06 — Tessellation and lattices. The same measurement-and-remap logic, applied to a connected system instead of loose components.

BUILD

P2b is due Week 9. Get all three attractor types running on the same array — the comparison is what is being graded.

BRING

To the midterm: the grayscale field and a first geometric response, as work in progress.

FINISH

P2a and P1c are both due at the midterm, along with the GSM. Do not leave the print to the last week — the queue is shared.